

EVIDENCE FOR BOSE CONDENSATION-LIKE EXCITATION OF COHERENT MODES IN BIOLOGICAL SYSTEMS

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Experimental evidence of the action of coherent millimeter electromagnetic radiation on biological systems shows very striking effects which are strongly frequency selective at very low power supply thus indicating resonance effects. The existence of a threshold for the power supply below which no effects are found strongly supports the earlier theoretical prediction that energy supply above a critical rate will lead to excitation of polar coherent modes in the system in a manner comparable to Bose condensation.

Some time ago it was suggested that many biological systems have a branch of polar modes in the $10^{11} - 10^{12} \text{ s}^{-1}$ frequency region, and that supply of energy to these modes leads to very strong (coherent) excitation of a single mode provided the rate of supply of energy, S , is larger than a critical one, S_0 , i.e. $S > S_0$, [1-3]. In the formal treatment the transition from $S < S_0$ to $S > S_0$ shows great similarity with Bose condensation.

The vibrations are considered to involve regions of biological membranes, or substantial sections of giant molecules — such as proteins, DNA, etc. Coherent excitation of these modes could have very far reaching biological influences in particular in the case that the excited mode represents a giant dipole oscillation (polarisation wave with zero wave number, $k = 0$). Amongst the many consequences of such excitations are the establishment of long range and short range selective forces [4] which so far have been invoked to interpret a number of specific biological features [5, 6], but which might in very many cases be the agent which initiates positional or conformational changes from which a variety of biological effects might arise (cf also [7]).

While the vibrations with higher wave numbers may be expected to be largely — though not exclusively — longitudinal no such classification holds for the giant dipole ($k = 0$) oscillation. Furthermore the linear dimension of most of the relevant biological objects (molecules, membranes) is less than the wavelength of electromagnetic radiation in the $10^{11} - 10^{12} \text{ s}^{-1}$ frequency region. It follows then

that radiation in resonance with the $k = 0$ frequency would be selectively absorbed by the material. The energy of excitation (cf [3] with minor modifications for the present case) would then be shared with other modes as long as $S < S_0$. Ultimately this would lead to an increase in temperature. For $S > S_0$, however, a single vibrational mode would be strongly (coherently) excited in a manner comparable to a Bose condensation. This mode may — but need not — be the mode $k = 0$ which absorbs the radiation; in this case S_0 would be relatively small.

Recently in a report of a meeting of the USSR Academy of Science (January 1973) [8] several remarkable contributions have demonstrated a striking influence of 6–7 mm coherent electromagnetic radiation on the behaviour of a variety of biological objects. The corresponding frequency of $\frac{1}{2} \times 10^{11} \text{ s}^{-1}$ is close to the theoretically predicted region for such effects. While the reported biological effects represent without doubt very complex processes they exhibit two very simple common features; (i) they depend very strongly on frequency exhibiting a typical resonance behaviour, and (ii) (cf. fig. 1 of [9] and fig. 3 of [10]) there exists a critical power supply $S = S_0$ below which no effects are observed and above which the observed effects are nearly independent of S .

These features strongly support the theoretical prediction of the existence of polar vibrations in the mentioned frequency region and of their biological importance when excited coherently. The crucial point here is the evidence for the existence of a threshold S_0 such that for $S > S_0$ a complete change of behaviour occurs comparable to a phase transition at $S = S_0$.

The theoretical interpretation of these results rests on the proposal that arising from the $k = 0$ resonance absorption the experimentally observed transition at $S = S_0$ consists in the coherent excitation of the polar mode mentioned above which does exhibit the required transition at $S = S_0$. This coherent excitation which is possible only when $S > S_0$ should then initiate the essential first step (e.g. selective attraction of molecules, opening of a section of a DNA complex, etc) in a possibly very complicated sequence of biological events. The time t_1 , required for this first step will decrease with increasing coherent excitation, i.e. with S (provided $S > S_0$), but the time t_2 for all the subsequent processes will be independent of S . An S value will thus be reached for which $t_1 \ll t_2$ holds so that for larger S the total rate will become nearly independent of S in agreement with the observations.

It thus appears that the very simple common features discovered in experiments on a variety of biological objects find their natural interpretation in the previous theoretical predictions. It should be emphasised that the new concepts introduced there make use of a greatly simplified model. These concepts are,

however, of a very general nature and permit the considerable refinements and modifications of the model which will be required for the detailed description of the observed phenomena. It is very striking to see, however, that complex biological phenomena may contain common features of very great simplicity in terms of the appropriate physical concepts.

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